

EXP. NO: 18 MATLAB SIMULATION AND EVALUATION OF EYE DIAGRAM PARAMETERS UNDER INTER-SYMBOL INTERFERENCE IN

DIGITAL COMMUNICATION SYSTEMS

Objectives

1. To simulate a digital communication system in MATLAB and generate the eye diagram of the received signal for evaluating eye height, eye width, eye-crossing level and timing margin.
2. To introduce and study Inter-Symbol Interference (ISI) under different channel conditions, and evaluate its effect on eye-diagram opening and overall digital signal quality.

CODE

```
clc;
clear;
close all;
% Parameters
N = 100;
sps = 20;
% Generate random binary data
data = randi([0 1], 1, N);
% NRZ encoding
tx = repelem(data, sps);
% Time axis
t = (0:length(tx)-1)/sps;
% Channel causing ISI
h = [0.2 0.6 1 0.6 0.2];
% Received signal
rx = conv(tx, h, 'same');
% Plot results
figure;
subplot(4,1,1);
stairs(data, 'LineWidth', 1.5);
grid on;
title('Original Binary Data');
xlabel('Bit Number');
ylabel('Bit');
ylim([-0.2 1.2]);
subplot(4,1,2);
```

```

plot(t, tx, 'LineWidth', 1.5);
grid on;
title('Transmitted NRZ Signal');
xlabel('Time (Bit Periods)');
ylabel('Amplitude');
ylim([-0.5 1.5]);
subplot(4,1,3);
plot(t, rx, 'r', 'LineWidth', 1.2);
grid on;
title('Received Signal with ISI');
xlabel('Time (Bit Periods)');
ylabel('Amplitude');
subplot(4,1,4);
eyediagram(rx, 2*sps);
title('Eye Diagram under ISI');

```

OUTPUT

